



2.14

Logarithmic Function Context and Data Modeling

Student Notes and Guided Practice

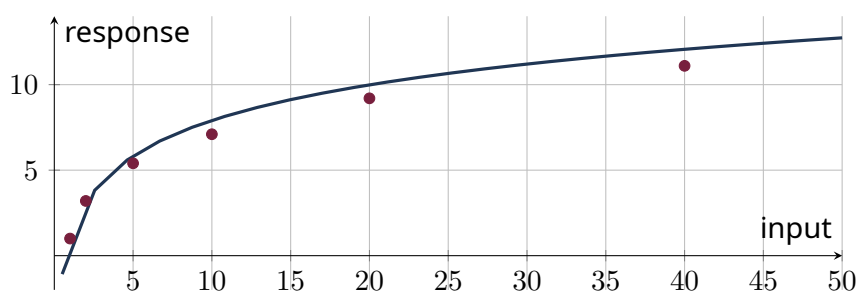
AP Precalculus - Unit 2 | ECHS Mathematics | Teacher: Mohammad Abu-Ghuwaleh

Learning Targets

- Construct logarithmic models from transformations, two points, and regression.
- Interpret logarithmic parameters and domains in context.
- Use models for forward and inverse prediction.

Essential Question

When does a logarithmic model appropriately describe rapid early change that slows as input grows?



Concept Framework

Key Idea

A logarithmic model often fits responses that change by equal amounts when input is multiplied by equal factors.

Key Idea

The domain must keep the logarithm argument positive.

Key Idea

Two data points determine scale and shift once a base and horizontal shift are chosen.

Key Idea

Extrapolation toward the vertical asymptote or far beyond data can be especially unstable.

Formula Map

Formula and Structure

$$y = a + b \ln(x - h)$$

Formula and Structure

$$y = a + b \log_{10}(x - h)$$

Formula and Structure

$$x = h + e^{(y-a)/b}$$

Worked Examples

Worked Example 1

Problem. Find $y = a + b \ln x$ through $(1, 2)$ and $(e^2, 8)$.

Solution. At $x=1$, $a=2$. Then $8 = 2 + 2b$, so $b = 3$. Model: $y = 2 + 3 \ln x$.

Worked Example 2

Problem. Solve $12 = 4 + 2 \ln x$.

Solution. $\ln x = 4$, so $x = e^4$.

Worked Example 3

Problem. Give a contextual limitation of a sound-level logarithmic model.

Solution. The input intensity must be positive and measurements outside the calibrated range may not follow the same relationship.

Multiple-Representation Connection

AP Reasoning

For this topic, always connect at least two representations: algebraic structure, numerical evidence, graphical behavior, or contextual meaning. State restrictions and units before interpreting a result. A correct calculation without a valid domain or contextual interpretation is incomplete AP reasoning.

Common Errors to Avoid

Common Error Check

- Do not discard domain restrictions when rewriting an expression.
- Do not infer local behavior from only one interval-wide average.
- Do not report a numerical answer without units or contextual meaning when quantities are named.
- Do not use a graphing window as proof that a feature does not exist.

Guided Check

1. Construct or use the logarithmic model: through (1,3),(e,7) in form $a+b \ln x$.

2. Construct or use the logarithmic model: VA $x=2$, point (3,5), form $\ln(x-2)+k$.

3. Construct or use the logarithmic model: change +6 when x multiplied by 10, $f(1)=2$.

4. Construct or use the logarithmic model: through (1,0),(100,8) base 10.

5. Construct or use the logarithmic model: model $y=5+2\ln x$ solve $y=11$.

6. A response increases 3 units whenever input doubles and equals 10 at $x=4$. Construct a base-2 model.

Lesson Summary

Complete these sentences before leaving the lesson:

1. The most important structure in this topic is _____.
2. A restriction I must remember is _____.
3. One graphical feature I can justify algebraically is _____.